

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING APPROACHES FOR AVIATION CYBERSECURITY: AN OVERVIEW

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Abstract

Artificial Intelligence (AI) and Machine Learning (ML) applications are currently found across all engineering domains, including cybersecurity engineering. This paper presents an overview of the AI-systems and technologies that can greatly benefit the security domain and its potential solutions for the aviation industry: anomaly detection for avionics, securing data link communications, and security certification, among others.

As showcased in the “National Strategy for Aviation Security” report in December 2018 and the U.S. Government Accountability Office report “Aviation Cybersecurity” in October 2020, there is a need for resilient cybersecurity practices and approaches to address the current issues that aviation faces today. Designing and implementing solutions to address these issues without exploring the feasibility of harnessing AI-powered cybersecurity tools would overlook the potential advantages these technologies can offer.

Therefore, this paper aims to provide a roadmap to adapt well-known ML-cybersecurity approaches to aviation security engineering and airworthiness: (i) autonomous and semi-autonomous cybersecurity for autonomous flight operations security, (ii) game theory models for adversarial and uncertainty modeling, (iii) human-AI interfaces for airport security monitoring and decision-making assistance, (iv) predictive analytics for anomaly detection for avionics and e-enabled aircraft, and (v) AI-based reasoning trustworthiness for software reliability and security certification. This paper also presents the challenges of including AI-cybersecurity in the aviation ecosystem to ground the proposed solutions within an accepted set of industry regulations, such as design verification for AI/ML algorithms and certification specifications for AI/ML solutions for manufacturers and agencies.

Introduction

Generally, it is understood that cybersecurity within the aviation ecosystem is a global threat that requires a global solution. The International Civil Aviation Organization (ICAO) recommended that each Contracting State should develop measures to address cybersecurity [1].

In the United States, the National Strategy for Aviation Security published December 2018 [2], aimed to address the enhancement of the safety and security of the aviation ecosystem while preserving the freedom of operations. Its objectives were described as follows: (i) protect the US and Global Interest in the Aviation Ecosystem, (ii) maximize aviation ecosystem security while maintaining aviation safety and balancing US economic impact, (iii) enhance resilience, mitigate damage, and expediate recovery, and (iv) effectively engage international, domestic, and private sector partners.

Its strategic actions were clearly stated: (i) maximize domain awareness, (ii) anticipate threats and assess vulnerability to and from the aviation ecosystem, (iii) ensure continuity and promote resilience of the aviation domain, and (iv) enhance international cooperation [3].

While aviation cybersecurity research is still ongoing, studying AI/ML algorithms seem to be the next step for cybersecurity engineering (as it has been shown for other industries). Therefore, the main goal of this paper is to address the following question: “*Can aviation benefit from AI-based cybersecurity?*”

Artificial Intelligence and Machine Learning for Cybersecurity

Artificial Intelligence (AI) and Machine Learning (M.L.) methodologies and techniques have been surveyed [4] and reviewed at length [5] during the past years. While the actual description of the common AI/ML methodologies and techniques is out

of scope of this paper, a brief classification based on the nature of the learning would be [6]:

- **Supervised learning:** the system has inputs and desired outputs which are known (labeled), with the purpose of mapping them. The most common techniques are *regression* and *classification*, which need labeled datasets for their training and testing.
- **Unsupervised learning:** the system finds patterns between inputs but with no (apparent) desired outputs. The most common technique is *clustering*, which does not need labeled data to extract patterns.
- **Reinforced learning:** the system is provided feedback and a set of rules. *Deep learning* and *associative reinforced learning* are some of the common techniques.

A.I. and M.L. Classification for Cybersecurity

AI/ML methodologies can also be classified by which types of applications they are part of. A five-type general description of AI/ML methodologies for cybersecurity is presented in Figure 1.

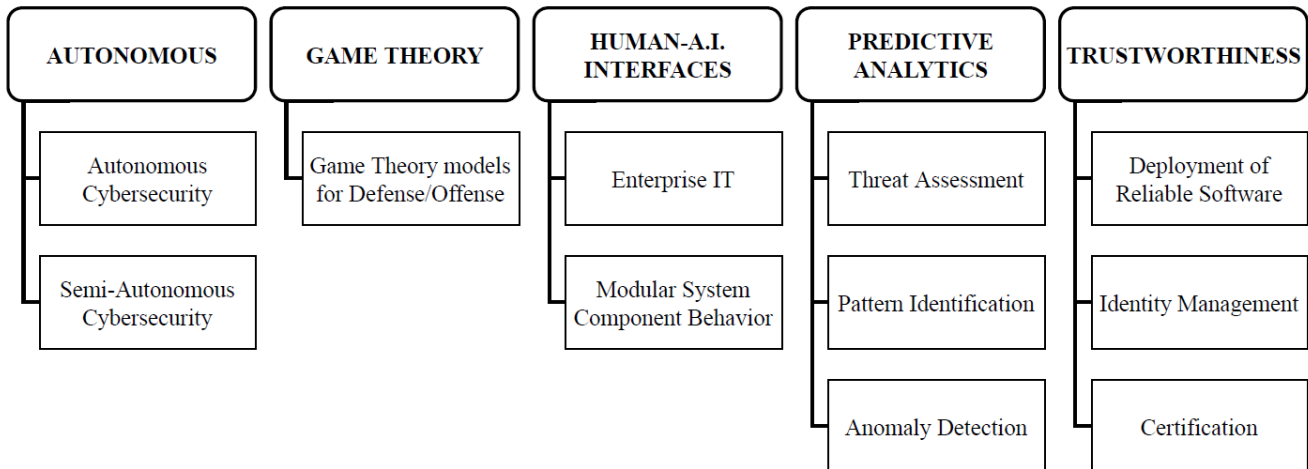


Figure 1. Types of A.I. and M.L. for Cybersecurity Engineering

Game Theory

The AI/ML methodologies are used to model any cyber defense of cyber offense capabilities and scenarios.

Autonomous Cybersecurity

The AI/ML methodologies are used for autonomous cybersecurity (which is independent from human interaction) and semi-autonomous cybersecurity (which contains semi-independent and human-in-the-loop paradigms).

In this category, the AI/ML methods are used to plan for the worst-case scenarios and analyze potential and actual threats. AI is also used to generate attack plans and goals, as well as actions that can help discover an attacker's intention.

- *Challenges:* due to the nature of the automation, avoiding conflicting security measures should be a priority.
- *Research:* AI/ML can be used to make systems resistant to autonomous analysis or autonomous attacks with no human interaction; it can also assist in determine the attacker's intent. There is also a need to translate cybersecurity mission statements and operations into inputs for automated cybersecurity.

The AI/ML methods are used to simulate and show differential scenarios for cybersecurity. The scenario takes into account changes in environmental values, player incentives, and other attack discoveries, and produces a new output accordingly.

Some scenarios can plan for irrational players that would act in unpredictable ways.

- *Challenges:* cooperation between several AI systems and teams.
- *Research:* AI/ML methods for additional AI techniques for uncertainty planning, in cooperative, as well as noncooperative environments; for assumptions in attacker capabilities and incentives. Further research into the game's action space is also needed.

Human-A.I. Interfaces

In this case, the AI/ML methods are focused on the coordination and trust between AI-cybersecurity systems and the people that interact with them.

These AI/ML methodologies are used to determine the best way to team with human resources to maximize the positive outcomes of the system and minimize the negative consequences. The AI system objectives need to be honored by all individual system components, so that they do not maximize their own goals at the expense of the system's.

- *Challenges:* Hybrid good decision making for human and AI. Information can be manipulated to the point that a person could trust an AI-generated video/voice recording. Decision-making assistance from AI and predictive models are needed when humans are not able to make the correct decisions effectively.
- *Research:* The research should be focused on human-machine teaming, AI-decision making assistance, and AI models that can predict human erroneous decisions.

Predictive Analytics

The most common type of AI/ML methods and techniques are predictive analytics, which take advantage of a *priori* knowledge, i.e., classified sources. They process information to assess the likelihood of an attack and discover patterns, links, and anomalies of different datasets from human/systems domains.

In these AI/ML methods the information from internal and external sources can be cross-referenced:

new patterns might be discovered, which a traditional analysis might not be able to unveil. Cyber and human domains can also be cross-reference in search for patterns that might not be apparent.

- *Challenges:* obtaining the clean, labeled, and real data; or, in its absence, synthetic but realistic data for training the AI/ML algorithms.
- *Research:* potential research in building datasets with real data or synthetic data that portray an accurate representation of reality. Also, research in validation, cross-referencing, and identifying potential flaws of existing datasets.

Trustworthiness

The AI/ML methodologies are employed to allow systems to become more trustworthy with AI-based reasoning when the right cybersecurity policies are used.

The AI/ML methods are used to securely develop, deploy, and operate software systems. It can detect low-level attack vectors of logic errors that traditional analysis might not. AI systems can also categorize several attacks and produce adaptive responses when inconsistencies are found across the systems. AI-based tools can be used as access control tools based on expected behavior instead of a token.

- *Challenges:* ethical and privacy violations arise when there is monitoring of behavioral patterns for identity management.
- *Research:* the AI/ML research is focused on software development and the trustworthiness of the AI system itself.

Aviation Cybersecurity

Aviation cybersecurity requires an in-depth knowledge of the aviation ecosystem, as well as industry-specific policies, specifications, and regulations. As a general description, one could classify aviation cybersecurity in four distinctive branches when addressing the aircraft as focus:

- **Aircraft as a System of Systems:** the aircraft is understood as a system-of-systems box. Common security issues include:

avionics security [7], internal networking security [8], COTS on board, etc.

- **Aircraft as a Node of a Network:** the aircraft is considered as one node of a large network of other aircraft and ground stations and satellites [9] [10] (e.g., aircraft mobility and security paradigms, ATN/IPS security).
- **Aircraft interaction within its Environment:** the aircraft is observed as a box that interacts with its environment using communications techniques. Its main security issues are studied in communication protocols such as ADS-B, ADS-C, CPDLC, ACARS, and AeroMACS.
- **Policies regarding the Aviation Ecosystem:** the regulations, standards, best practices [11], and security specifications that determine the aircraft design and operation and its aviation ecosystem.

A.I. and M.L. Approaches in Aviation Cybersecurity

There have been AI/ML reported advances in the aviation industry that are not related to cybersecurity [12] [13]. There are several other

aviation-related and aerospace areas that already benefit from AI [14-18]:

- Airline Performance: fuel usage optimization; improving routes.
- Automation and Autonomy: speech recognition models for controller assistance.
- Capabilities: environmental improvements; automatic speech recognition; automatic taxi, take-off, and landing enabled by computer vision.
- Passengers: biometrics for boarding.
- Resource management and optimization: ATM support/capacity balancing; optimization of capacity with current resources [19].
- Traffic prediction and forecast modeling: improving predictions.

Therefore, it is not difficult to imagine that aviation cybersecurity could benefit from AI/ML as well. Considering the initial classification of the AI/ML methodologies depending on application and the current branches of aviation cybersecurity, an adapted classification for AI/ML for aviation cybersecurity is presented in Figure 2.

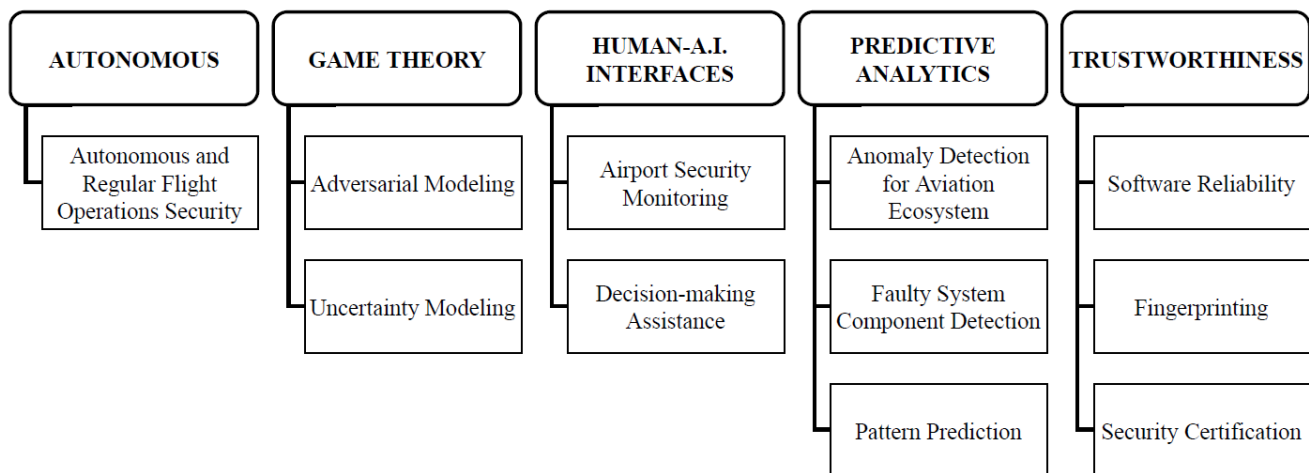


Figure 2. Adapted Types of A.I. and M.L. for Aviation Cybersecurity

Autonomous Cybersecurity

Autonomous cybersecurity AI/ML methodologies can be used for autonomous flight

and regular flight operational security. They can classify diverse types of events in Security Operations Center (SOC) anywhere in the aviation ecosystem [20], as well as provide autonomous decision aid for flight

operations [21]. They also help identify pattern conception that can later be used in predictive analysis; and offer support in risk analysis methodologies.

Game Theory

Game Theory AI/ML methodologies address adversarial and uncertainty modeling. They can improve the simulator capabilities for flight, as well as create more realistic threat scenarios for cyber defense/offense table-top exercises.

Human-AI Interfaces

The Human-AI interfaces AI/ML methods can be employed in airport security monitoring and provide decision-making assistance for human-in-the-loop systems. These AI/ML methods allow for: (i) adapting risk mitigation and security controls to needs of aviation domain [22], (ii) auto generation of attack trees for critical infrastructure and aviation systems of systems [23] [24], (iii) linking weak signals and modeling suspicious interactions inside networked systems or Air Traffic Operations systems data flows, (iv) automation for non-safety systems.

Predictive Analytics

Predictive Analytics are the most commonly studied AI/ML methods in aviation cybersecurity. They allow for anomaly detection for the aviation ecosystem and faulty system component detection in scenarios such as: (i) avoiding missing relevant information/cross-reference information [25], (ii) cyber monitoring and detection in aviation [26], (iii) intrusion detection [27], (iv) predicting cyberattacks and risk [28], (v) anomaly detection through component behavior [29] [30], (vi) isolate the attack and minimize its impacts.

Trustworthiness

Trustworthiness AI/ML methods in aviation cybersecurity can be used for software reliability and its deployment, as well as fingerprinting applications, and security certification. These methods can allow for: (i) safety-critical software development with AI testing/verification to identify low-level attack vectors, (ii) cross-reference attack

vectors with hardware and software for safety-critical systems.

A.I. and M.L. Challenges

As mentioned in Section II, AI/ML methodologies for cybersecurity do have some challenges to be fully useful. In a 2018 comparative study of AI/ML techniques for aviation [6], the main challenge highlighted is the unavailability of high-quality data due to the fact that most data are considered private or belongs to an industrial organization. This leads to a slow process of research and implementation of AI/ML models since the models themselves cannot be train/tested on real data. In addition, in [31] the Artificial Neural Networks (ANN) developed during the research stage could be assumed to work on an aviation application, but a challenge presented itself when one wanted to efficiently integrate a traditional airborne computer with a system with ANN, due to the lack of system verification.

Challenges

The following challenges need to be addressed to be able to fully develop AI/ML methodologies and frameworks for aviation cybersecurity.

Challenge #1: The Lack of Testbeds

Building testbeds for aviation applications is a costly endeavor that cannot always be afforded by research groups or small companies. The aviation ecosystem and its data, at the technical level, is highly regulated and follows specifications that cannot be accessed by all. The testbeds have to generate accurate and labeled data that can be useful for AI/ML research.

Challenge #2: No Readily Available Datasets

Since not everyone can build a testbed, readily available datasets would be useful to bring more research into the aviation cybersecurity focus. Unfortunately, this is not currently the case, as the datasets that could be used for aviation cybersecurity are scarce and do not cover all aviation cybersecurity domains.

Challenge #3: Policies, politics, and regulations

Aviation cybersecurity policies and regulations are not always quick and straightforward processes, and they are usually bound by politics from Contracting States (ICAO). There is a need for a civil aviation cybersecurity architecture [32] that can

address legislation and standards in a timely manner; and that can also regulate any upcoming AI/ML methodology that benefits aviation cybersecurity.

Roadmap for A.I. and M.L.

More research is needed to shift the AI/ML methods from a laboratory setting to a production environment. The E.U.A.S Agency [33] proposes an Artificial Intelligence Roadmap that follows a human-centric approach to develop AI/ML methodologies. While human-in-the-loop is a branch of cybersecurity engineering that benefits greatly from AI/ML methods, this paper proposes a general Roadmap that can be applied to any of the previously described types of AI/ML Aviation Cybersecurity in Figure 2.

The Roadmap (Figure 3) is divided into four smaller goals that are to be accomplished sequentially to ensure that all challenges are addressed and a robust AI/ML aviation security system is built:

- **Goal 1:** *Build testbed(s) for data gathering, dataset acquisition, and testing.*
 - End State: One or more testbeds are used to gather and modify different types of aviation data (correctly labeled) and to test different use cases of data.
- **Goal 2:** *Develop methodologies for aviation AI/ML algorithm robustness.*
 - End State: Methodologies for one or several of aviation AI applications are studied and implemented.
- **Goal 3:** *Build framework(s) for complex dataset AI/ML for training and testing.*
 - End State: One or more frameworks are designed and used to train/test different AI methods.
- **Goal 4:** *Develop new AI-based techniques for aviation cybersecurity.*
 - End State: New AI-based techniques are designed and implemented to address current and upcoming aviation cybersecurity challenges.

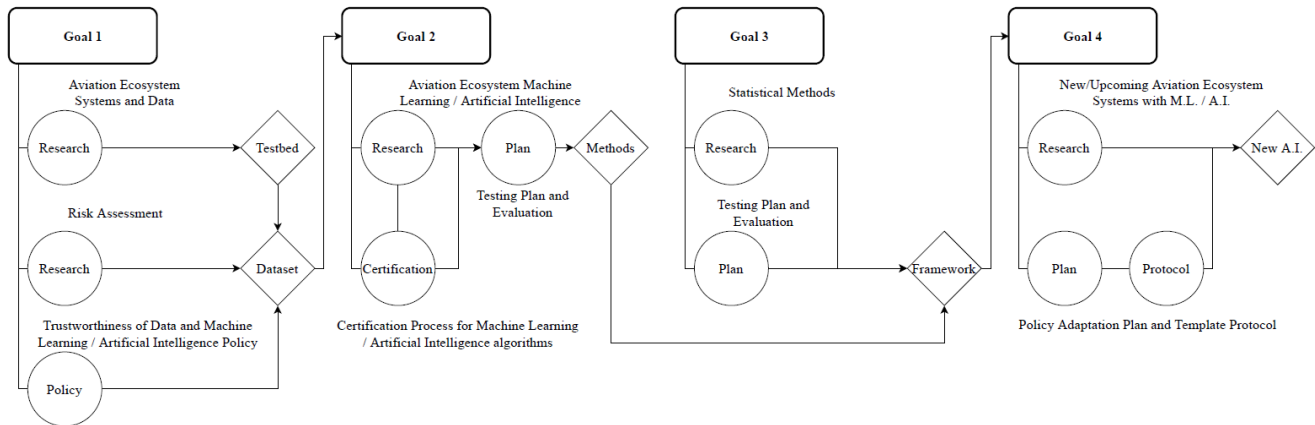


Figure 3. Roadmap for A.I. and M.L. for Cybersecurity in the Aviation Ecosystem

Conclusions and Future Research

Having reviewed the AI/ML and Aviation Cybersecurity current states, it can be concluded that the answer to the initial question “Can aviation cybersecurity benefit from AI/ML techniques?” is

affirmative, albeit it should consider the challenges outlined in this paper.

The state-of-the-art of AI/ML for cybersecurity is more advanced than the current state of AI/ML for aviation and combining both would require a joined effort from all states in ICAO. This, in turn, requires

an investment in research and developing new regulations, standards, and best practices for the AI/ML methodologies. Any research conducted should include: (i) a testbed and datasets with real and accurate data that reflects real-life situations, (ii) the creation and maintenance of realistic simulation environments, (iii) the development of avionics-specific datasets that are modular and can be shared to foster innovation and collaboration.

New regulatory actions should be developed to include the standards, best practices, and policies. The addition of AI/ML methodologies in the aviation ecosystem would affect future aircraft design, operations, maintenance, and any ATM operations, which would include different specifications than the ones that are currently in place. Additional Means of Compliance should be developed in the long run as well.

In conclusion, AI/ML methodologies and techniques can be used to increase aviation security, offering aid in decision making scenarios, finding patterns to determine risk, and detecting faulty components in the systems; however, there are still some steps ahead to make it fully operational.

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